Digital Forensics CTF Report Manual File Extraction from HTTP Traffic – PCAP

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Case File: bns_captured_activity.pcap

Tools Used: Wireshark, HxD Hex Editor

Introduction

The forensic CTF exercise focused on analysing a captured network session to manually extract files transferred over HTTP. The goal was to identify, reconstruct, and recover hidden files within TCP streams without relying on automation or scripting. Using only Wireshark and the HxD hex editor, I examined raw packet data to uncover transferred content and rebuild multiple file types, mirroring real-world scenarios where manual evidence recovery is essential.

Objectives

The goals of this analysis were to investigate HTTP traffic within the packet capture, follow TCP streams to locate file transfers, identify file types using hexadecimal signatures, manually extract and reconstruct .jpg, .pdf, and .zip files, .png and document all findings/ lesson learnt in a structured forensic report.

1. Analysis:

Traffic Analysis in Wireshark

The bns_captured_activity.pcap file was opened in Wireshark and HTTP-focused filters including http,

http.request.method == "GET" and

http.request.method == "POST"

were applied to isolate relevant traffic. These filters highlighted HTTP sessions where data were being transferred between hosts.

1.1 Following TCP Streams

Suspicious packets were investigated by selecting

Follow → TCP Stream

which allowed me to view full conversations and spotted the embedded file content within the data payloads.

1.2 Identifying File Signatures

Within each TCP stream, I searched for known hexadecimal file headers to confirm file types:

- JPG: FF D8 FF

- PDF: 25 50 44 46

- ZIP: 50 4B 03 04

- PNG: 89 50 4E 47

These signatures verified the presence and type of each transferred file.

1.3 Manual File Reconstruction

Once file content was identified, the raw hexadecimal data was copied from Wireshark, pasted into HxD, cleaned to remove unrelated stream data and saved with the correct file extension. Each reconstructed file was then opened to verify its integrity.

2. Recovered Evidence

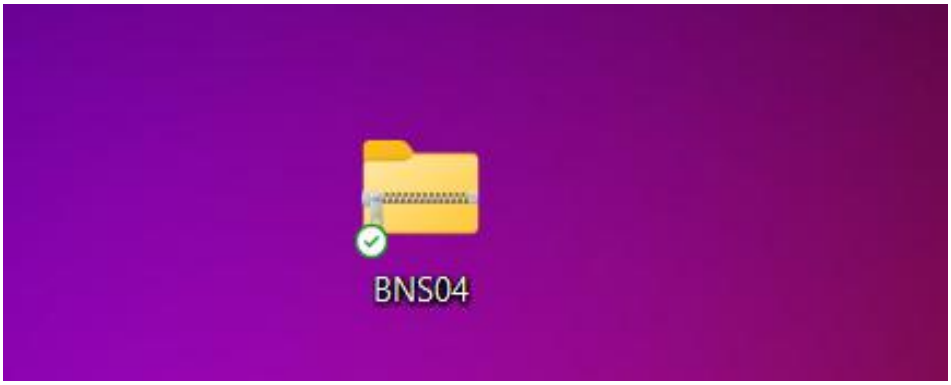
The following files were successfully extracted and reconstructed:

- BNS01.jpg — JPEG image — recovered

- BNS02.png — PNG image — recovered

- BNS03.pdf — PDF document — recovered

- BNS04.zip — ZIP archive — recovered



- Conclusion

  The captured traffic showed evidence of unauthorized file transfers hidden within normal HTTP communication. Through embedding files inside TCP streams, the activity attempted to blend in as normal routine web traffic. Manual extraction confirmed several unencrypted file types being transferred, underscore potential data-exfiltration techniques and demonstrated the value of hex-level analysis in exposing concealed material. This approach aligns with methods commonly seen in malware delivery, insider data threat and other - stealth communication methods.

  This analysis recovered four hidden files from HTTP traffic without relying on automation. The analysis - strengthened key network-forensic skills, including traffic investigation, TCP stream reconstruction and data extraction through hexadecimal signatures. The findings show how attackers can embed data within normal-looking traffic and underscore the importance of deep packet analysis during incident response. The findings show how attackers can embed data within normal-looking traffic and emphasise the need for deep packet analysis during incident response.